TMA3084/TMA3083 Software Engineering Laboratories
Project Evaluation Rubrics (Design & Implementation)
(25%)

These assignment rubrics are used to assess two deliverables, which are as follows:

Deliverables	Percentage
A. Requirement & Design Document	10
B. Implementation	15
TOTAL	**25**

	Section A: Requirement & Design Document (15 marks)	Section B: Implementation (10 marks)	Percentage (25%)
CLO1	**/15**		**/15**
CLO2		**/10**	**/10**

Section A: Requirement & Design Document (10 marks)

Criteria for assessment	*Performance levels*			
	Proficient (80 – 100%)	**Competent (50 – 79%)**	**Apprentice (20 – 49%)**	**Novice (0 – 19%)**
Problem Definition (10% Weighting)	Well-defined problem.	Adequately-defined problem.	Vaguely-defined problem.	No problem definition.
Requirements and Specification (15% Weighting)	Accurate, comprehensive, and sufficiently specific user requirements and technical specification.	Accurate user requirements and technical specification that cover most aspects of the system.	User requirements and technical specification cover only some aspects of the system, and miss some significant aspects, or characterize them accurately.	Insufficient user requirements and technical specification: meeting the stated requirements and specifications does not solve the stated problem.
System Architecture (25% Weighting)	1. Thorough description of the solution concept, algorithms, and alternative approaches. 2. All major system components are identified with appropriate abstraction. 3. Clear assignment of system functions to system components, covering all system functions. 4. Hardware vs. software components are identified. 5. Insightful discussion of alternative architectures and the involved tradeoffs, and convincing justification of architectural choices.	1. Reasonable description of the solution concept, algorithms, and alternative approaches. 2. Most major system components are identified, with mixed levels of abstraction. 3. Most system functions are assigned to specific system components. 4. Hardware vs. software components are identified. 5. Adequate discussion of alternative architectures, and adequate justification of architectural choices.	1. Incomplete description of the solution concept, algorithms, and alternative approaches. 2. Only some system components are identified. Some major components are missing. 3. Some main system functions are not mapped to any system components. 4. Unclear designation of hardware vs. software components. 5. Superficial discussion of alternative architectures. Unconvincing justification of architectural choices	1. No discussion of the general solution concept and algorithms. 2. Non-representative, or missing, list of abstract system components. 3. Unclear assignment of system functions to specific system components. 4. No alternative architectures are considered.

Component Design (15% Weighting)	1. Sound justification for off-the-shelf vs. custom components. 2. Off-the-shelf components: thorough comparison of alternatives; sound selection criteria. 3. Custom components: clear description of component design; all obvious design alternatives are considered.	1. Reasonable justification for off-the-shelf vs. custom components. 2. Off-the-shelf components: reasonable comparison of alternatives; biased selection criteria. 3. Custom components: reasonable description of component design; some design alternatives are considered, but some obvious ones are not.	1. Unconvincing or unsound justification for off-the-shelf vs. custom components. 2. Off-the-shelf components: basic comparison of alternatives; poor/missing selection criteria. 3. Custom components: incomplete description of component design; no design alternatives are considered.	1. No justification for off-the-shelf vs. custom components. 2. Off-the-shelf components: no alternatives are considered. 3. Custom components: no description of component design.
Mobile Application Integration (15% Weighting)	1. Inter-component interfaces are defined, with sound justification for custom vs. standard interfaces. 2. Custom interfaces are clearly specified. 3. Interactions between all interacting components are clearly described.	1. Inter-component interfaces are defined, with adequate justification for custom vs. standard interfaces. 2. Custom interfaces are adequately specified. 3. Interactions between most components are adequately described.	1. Inter-component interfaces are defined, but no justification for custom vs. standard interfaces. 2. Custom interfaces are not specified. 3. Interactions between some components are partially described.	1. Inter-component interfaces are not defined. 2. No discussion of interaction between application components.
Progress and Documentation (10% Weighting)	1. Made sufficient progress so far to complete the project in time. 2. Work is well-documented. Document paints a clear picture of the project progress.	1. Progressing slowly. Need to pick up the pace to complete the project in time. 2. Work is partially documented. Document is incomplete, and leaves many questions unanswered.	1. Noticeably behind schedule. Completing the project is questionable. 2. The document barely describes the project status.	1. Completing the project appears to be infeasible. 2. The document is unable to describe the project status.
Technical Writing (5% Weighting)	1. Well-structured document. 2. No grammar, spelling, or punctuation mistakes. 3. Clear presentation of ideas. 4. Excellent and complete background.	1. Well-structured document. 2. Few grammar, spelling, or punctuation mistakes. 3. Understandable presentation of ideas. 4. Reasonable background.	1. Awkward document structure. 2. Noticeable grammar, spelling, or punctuation mistakes. 3. Vague presentation of ideas. 4. Inadequate background.	1. Illogical document structure. 2. Frequent grammar, spelling, or punctuation mistakes. 3. Confusing presentation of ideas. 4. Required background missing.
Team work (5% Weighting)	1. The work load and variety on each member seems fair. 2. Leadership role being assumed by each member for different tasks is evident.	1. The work load and variety on each member seems fair. 2. Leadership role being assumed by each member for different tasks is NOT apparent.	The work load and variety on each member does not seem to be fair or at least one member is assigned trivial non-technical tasks (e.g. writing the report).	No teamwork: fewer than five members

Section B: Implementation
(15 marks)

Criteria for assessment	*Performance levels*			
	Proficient (80 – 100%)	**Competent (50 – 79%)**	**Apprentice (20 – 49%)**	**Novice (0 – 19%)**
Function implementation (50% weighting)	The program passes all the test cases and is implemented according to the requirements.	The program passes most test cases and is implemented according to the requirements or the program passes all the test cases but does not follow some of the requirements.	The program passes half around test cases and is implemented according to the requirements or the program passes all the test cases but does not follow requirements.	The program fails most cases but is implemented according to most requirements or the program passes most test cases but does not follow requirements.
AI API Integration (10% Weighting)	1. AI API integration is fully explained, including endpoint details, data flow, authentication method, request/response handling, and error-handling flow. 2. Clear justification of how the AI API enhances mobile functionality (e.g., chat, image generation, recommendation, classification). 3. Correct mapping between UI components, service layers, and AI endpoints (e.g., Provider, BLoC, Riverpod). 4. Proper handling of asynchronous calls, streaming responses, token limits, and rate limits. 5. Security considerations identified (API key protection, backend proxy, environment variables, secure storage). 6. Demonstrates understanding of performance impact (latency, caching, batching, offline fallback).	1. AI API integration described with some detail on endpoints and data flow. 2. Shows how the AI API contributes to app features, but explanation lacks depth. 3. UI-service-AI mapping present but incomplete. 4. Basic async handling is discussed; limited mention of edge cases. 5. Some security considerations included but not comprehensive.	1. AI API mentioned but described vaguely; limited details on endpoints or data structure. 2. Weak or unclear explanation of how the AI improves the app. 3. Poor linkage between API operations and UI or backend flow. 4. Minimal mention of async handling, errors, or API limits. 5. API key and security considerations missing or weak.	1. No meaningful AI API integration described. 2. AI is mentioned only at idea level; no technical explanation. 3. No mapping between API, UI, and system flow. 4. No consideration of security, latency, or API behavior.

Program structure (20% weighting)	The program structure follows design completely.	The program structure follows most part of the design.	The program structure follows only a few parts in the design.	The program structure follows very few parts in the design.
Comment (10 % weighting)	Comments are appropriately added and explain the program very well.	Comments help the explanation but are improper at some places.	Comments are provided with some help in program understanding.	Comments are sparse or vague.
Code style (10 % weighting)	A clear coding style is consistently applied, greatly enhancing program readability.	A clear coding style is consistently used in the most part of the program, helping in the program readability.	A coding style is used in a small part of program. The program is not readable at some places.	Only a very few efforts are made in improving the